

Supplementary Material: Search Procedure for the Tie-Manifold Case Study

BSD (bsd.developer@proton.me)
Independent Researcher

Draft updated 2 October 2026

Search procedure and result boundary

This supplement accompanies *A Machine-Checked Level-Four Bound on the Matrix Multiplication Exponent: A Tie-Manifold Search Case Study*. AI agents assisted under the author’s direction (paper, Appendix D). The procedure below is a retrospective synthesis, not the verbatim initial prompt or a guarantee of reproducing the numerical trajectory. The private optimizer and campaign records are not prerequisites for checking the supplied certificate.

The earlier eight-hour campaign began at numerical score 2.3711894606957515 and ended at 2.371176825455252. Subsequent continuation selected **point H**, with numerical score 2.371174031813689. The certified result is instead

$$\omega_{\mathbb{Q}} \leq \frac{43740440563687076465}{2^{64}} = 2.37117403423113 \dots$$

The paper measures branch geometry at point H using its frozen source. The earlier prediction-agreement example concerns an earlier campaign model step.

Retrospective research procedure

1. **Establish a numerical baseline.** Use a pinned implementation and an independently validated physical checkpoint. Distinguish the current global incumbent from a restored point used only to construct a local model. Preserve the best physical point even when a better-balanced construction seed has a worse hard score.
2. **Diagnose branch geometry.** Measure branch values, forty contrasts, their Jacobian, fitted convex weights and the projected gradient at the actual model origin. Check numerical rank and positivity. Use the exact mixture/slack identity to distinguish surrogate gain from hard gain; conditional search lemmas do not prove floating-point observations.
3. **Construct a reduced tangent model.** Form coupled tangent directions and estimate curvature using first-gradient differences at two scales. Check scalar slopes, mixed symmetry and model positivity. The native double-backward path failed a proved zero-curvature diagnostic and was not used for these models. Keep retained metrics associated with their original construction points.
4. **Evaluate finite moves and corrections.** Test bounded fractions with and without balance restoration. Target zero contrasts only where the branch and convex-mixture conditions justify it. Respect every physical domain. Select using the original hard exponent, never model gain alone. Record predictions, realized gains, slack and correction effects.
5. **Compare matched alternatives.** Compare model widths or alternative methods from the same checkpoint under explicit finite-trial controls, including setup costs. Added directions can improve the score while failing to justify their computation cost. Short unsuccessful controls do not establish general optimizer inferiority.
6. **Validate each promotion.** Require production, independently implemented reference and serialized/reloaded readouts. The recorded configuration was native400/reference1600/native400. Independently reconstruct the move and score components. A numerical crossing is not an exact bound or a convergence certificate.

7. **Certify the selected endpoint separately.** Freeze the rationalized witness, construct maximum-entropy witnesses and directed enclosures, and check the complete canonical bytes. Require independent exact Rust agreement, full Lean evaluation and compiled statement/axiom auditing. Distinguish the numerical score, exact claim and every remaining trust assumption. Search diagnostics are not premises of the bound theorem.

What the reader can verify

The verification package `certifications/l4-2026-10-01-best015` contains the matching checker sources, dependency locks, assurance and trust records. Its `verification-runbook.md` separates file-identity checking, source compilation, independent exact Rust checking and full Lean CN certification. The canonical certificate is 458,921,565 bytes with SHA-256

```
dfcede72c9dbf8eaf7b55f34cdd56d93dca7fe8a28fca2787466bec410f1873f
```

The current result depends on `Classical.choice`, `propext`, `Quot.sound`, one certificate-specific native-evaluation axiom, and `MatrixMath.AX1_combination_loss`, together with the relevant compiler, runtime and GMP trust. Removing native evaluation and proving the full combination-loss bridge are ongoing work, not completed results.

Two full local CN executions and their audits passed. Local retrieval and byte comparisons do not establish completion of the separate remote release-verification protocol. The certificate establishes one endpoint under the disclosed assumptions; it does not prove global optimality, search reproducibility or a comparative speed advantage.